The Culinary Fingerprint: Hub, Community, and Bridge Analysis of Statistically Validated Ingredient Backbones

Methodology and Scope

This report characterizes the descriptive ingredient co-occurrence backbones constructed from all 17,279 recipes across Indian (3,003), Italian (7,838), and Mexican (6,438) cuisines in the Kaggle What's Cooking dataset (Table 1). Note that descriptive networks use the full selected dataset, whereas predictive model artifacts in Deliverable 4 are rebuilt exclusively on the 80% stratified training split to prevent data leakage. For each cuisine with N recipes, binary recipe-level ingredient presence defines marginal probabilities P(a) = n(a)/N and joint probabilities P(a,b) = n(a,b)/N among eligible ingredients occurring in at least 20 recipes (Table 1). Pointwise Mutual Information is defined as PMI(a,b) = log2(P(a,b) / (P(a)P(b))), and Normalized Pointwise Mutual Information is NPMI(a,b) = PMI(a,b) / (-log2 P(a,b)) in [-1, 1] (with limiting convention NPMI = 1 when P(a,b) = 1). To validate edges against degree heterogeneity, one Curveball Markov chain (Strona et al., 2014) is run per cuisine on the full binary recipe-by-ingredient incidence matrix, preserving every recipe's exact ingredient count and every ingredient's exact recipe frequency across 1,000 samples after a burn-in of 10 x N trades and thinning of 2 x N trades per sample (Table 1). Each pair's null mean mu_null and sample standard deviation sigma_null yield z(a,b) = (n(a,b) - mu_null) / sigma_null; pairs with sigma_null = 0 are conservatively excluded, and the statistically validated backbone retains edges satisfying z(a,b) > 3 (strict inequality; an uncorrected single-pair threshold) and co_count >= 5 (Table 1, Figure 1). On each undirected backbone, an ingredient's weighted degree (strength) is Strength(a) = sum of max(NPMI(a,b), 0) over incident backbone edges, normalized weighted betweenness centrality is computed over shortest paths with edge distance d(a,b) = 1 / max(NPMI(a,b), 0) (excluding zero-strength edges), and structural bridge importance between cuisines A and B for shared backbone nodes is Bridge(a; A,B) = min(Strength_A(a), Strength_B(a)) (Tables 2-7, Figures 2-5). In Deliverable 4, the training-only Fingerprint Scorer sums backbone NPMI over matched recipe pairs, Score(c) = sum_{(a,b) in M_c(R)} NPMI_c(a,b), whereas its normalized ablation computes the matched-pair mean Score_norm(c) = Score(c) / |M_c(R)| (0 if |M_c(R)| = 0); empirically, dividing by |M_c(R)| achieves substantially lower held-out accuracy than summing, consistent with the structural property that averaging over |M_c(R)| removes cumulative pair-count evidence and allows a single matched edge in one backbone to exceed the multi-edge mean in another.

Table 1: Dataset, Null-Model Mixing, and Validated Backbone Summary

Cuisine	Recipes	Distinct Ing.	Eligible (>=20)	Backbone Nodes	Edges Before	Edges After	Retained (%)	Communities	Modularity	Null ESS(S)
Indian	3,003	1,659	268	261	22,284	1,829	8.21%	7	0.488	655
Italian	7,838	2,924	487	473	48,488	3,758	7.75%	6	0.473	667
Mexican	6,438	2,676	423	408	42,703	2,847	6.67%	7	0.497	691

(a) Top-10 hubs

In the Indian backbone (261 nodes and 1,829 validated edges, retaining 8.21% of 22,284 raw edges; Table 1, Figure 1), the top three weighted-degree hubs are clove (strength 12.77, rank 1, betweenness 0.0170 [rank 38], prevalence 10.19%), garlic paste (strength 12.28, rank 2, betweenness rank 20, prevalence 7.36%), and green cardamom (strength 12.18, rank 3, betweenness rank 17, prevalence 2.56%) (Table 2, Figure 2). By weighted betweenness centrality (distance 1/max(NPMI,0)), the highest-ranked brokers in Indian are peeled fresh ginger (betweenness 0.0540, rank 1; strength 8.60, rank 13) and heavy cream (betweenness 0.0397, rank 2; strength 6.79, rank 25), with no ingredients appearing in the top 10 on both metrics (Table 2, Figure 2).

In the Italian backbone (473 nodes and 3,758 validated edges, retaining 7.75% of 48,488 raw edges; Table 1, Figure 1), the top three weighted-degree hubs are vanilla extract (strength 18.24, rank 1, betweenness 0.0235 [rank 15], prevalence 2.79%), carrot (strength 18.11, rank 2, betweenness rank 1, prevalence 4.84%), and sugar (strength 15.50, rank 3, betweenness rank 32, prevalence 9.70%) (Table 3, Figure 2). By weighted betweenness centrality (distance 1/max(NPMI,0)), the highest-ranked brokers in Italian are carrot (betweenness 0.0527, rank 1; strength 18.11, rank 2) and cooking spray (betweenness 0.0466, rank 2; strength 13.69, rank 9), with carrot, baking powder, cooking spray appearing in the top 10 on both metrics (Table 3, Figure 2).

In the Mexican backbone (408 nodes and 2,847 validated edges, retaining 6.67% of 42,703 raw edges; Table 1, Figure 1), the top three weighted-degree hubs are ancho chile pepper (strength 13.80, rank 1, betweenness 0.0387 [rank 3], prevalence 2.07%), ground cinnamon (strength 13.04, rank 2, betweenness rank 5, prevalence 2.73%), and dried oregano (strength 11.51, rank 3, betweenness rank 52, prevalence 8.05%) (Table 4, Figure 2). By weighted betweenness centrality (distance 1/max(NPMI,0)), the highest-ranked brokers in Mexican are cooking spray (betweenness 0.0460, rank 1; strength 7.85, rank 30) and baking powder (betweenness 0.0425, rank 2; strength 11.22, rank 6), with ancho chile pepper, ground cinnamon, baking powder appearing in the top 10 on both metrics (Table 4, Figure 2).

Table 2: Top-10 Weighted-Degree and Top-10 Weighted-Betweenness Hubs in Indian

Ingredient	Strength	Str. Rank	Betweenness	Btw. Rank	Deg.	Prev. (%)	Top-10 Str.	Top-10 Btw.
clove	12.77	1	0.0170	38	48	10.19%	Yes	No
garlic paste	12.28	2	0.0227	20	47	7.36%	Yes	No
green cardamom	12.18	3	0.0235	17	36	2.56%	Yes	No
mace	10.75	4	0.0245	14	28	1.43%	Yes	No
green chilies	10.44	5	0.0228	19	51	16.12%	Yes	No
oil	10.36	6	0.0196	31	50	18.18%	Yes	No
red chili powder	10.22	7	0.0103	71	41	5.69%	Yes	No
cinnamon sticks	9.97	8	0.0225	22	36	8.52%	Yes	No
shahi jeera	9.55	9	0.0226	21	22	0.67%	Yes	No
bay leaf	9.54	10	0.0103	72	34	3.93%	Yes	No
peeled fresh ginger	8.60	13	0.0540	1	32	5.16%	No	Yes
heavy cream	6.79	25	0.0397	2	28	3.50%	No	Yes
active dry yeast	5.55	45	0.0341	3	15	1.13%	No	Yes
curry leaves	7.16	22	0.0308	4	29	6.93%	No	Yes
ground cardamom	4.68	63	0.0304	5	19	5.79%	No	Yes
ground red pepper	6.00	35	0.0297	6	22	3.00%	No	Yes
boneless chicken skinless thigh	4.72	62	0.0294	7	21	2.06%	No	Yes
green cardamom pods	4.41	75	0.0291	8	15	2.26%	No	Yes
saffron	6.50	29	0.0284	9	23	2.50%	No	Yes
large garlic cloves	4.05	84	0.0268	10	15	1.96%	No	Yes

Table 3: Top-10 Weighted-Degree and Top-10 Weighted-Betweenness Hubs in Italian

Ingredient	Strength	Str. Rank	Betweenness	Btw. Rank	Deg.	Prev. (%)	Top-10 Str.	Top-10 Btw.
vanilla extract	18.24	1	0.0235	15	59	2.79%	Yes	No
carrot	18.11	2	0.0527	1	83	4.84%	Yes	Yes
sugar	15.50	3	0.0148	32	66	9.70%	Yes	No
tomato paste	14.90	4	0.0243	12	68	4.80%	Yes	No
all-purpose flour	14.82	5	0.0175	24	68	11.71%	Yes	No
onion	14.80	6	0.0130	44	91	15.82%	Yes	No
baking powder	14.53	7	0.0263	10	47	2.37%	Yes	Yes

Ingredient	Strength	Str. Rank	Betweenness	Btw. Rank	Deg.	Prev. (%)	Top-10 Str.	Top-10 Btw.
dry white wine	14.01	8	0.0237	13	76	8.39%	Yes	No
cooking spray	13.69	9	0.0466	2	64	6.26%	Yes	Yes
large egg	12.91	10	0.0149	31	63	7.97%	Yes	No
arborio rice	11.55	17	0.0426	3	46	3.33%	No	Yes
ground pork	9.75	32	0.0383	4	37	0.92%	No	Yes
fresh parmesan cheese	9.99	29	0.0312	5	43	3.20%	No	Yes
ground beef	11.93	15	0.0301	6	51	3.02%	No	Yes
celery ribs	10.26	26	0.0290	7	42	1.66%	No	Yes
caper	7.55	56	0.0268	8	36	3.90%	No	Yes
part-skim mozzarella cheese	9.63	36	0.0265	9	37	2.56%	No	Yes

Table 4: Top-10 Weighted-Degree and Top-10 Weighted-Betweenness Hubs in Mexican

Ingredient	Strength	Str. Rank	Betweenness	Btw. Rank	Deg.	Prev. (%)	Top-10 Str.	Top-10 Btw.
ancho chile pepper	13.80	1	0.0387	3	46	2.07%	Yes	Yes
ground cinnamon	13.04	2	0.0370	5	45	2.73%	Yes	Yes
dried oregano	11.51	3	0.0121	52	66	8.05%	Yes	No
all-purpose flour	11.46	4	0.0159	31	41	5.13%	Yes	No
garlic cloves	11.43	5	0.0088	88	70	14.31%	Yes	No
baking powder	11.22	6	0.0425	2	34	2.66%	Yes	Yes
cumin	10.30	7	0.0259	12	56	9.71%	Yes	No
ground cumin	9.99	8	0.0082	95	67	20.91%	Yes	No
sugar	9.95	9	0.0225	17	41	5.33%	Yes	No
clove	9.60	10	0.0066	116	27	0.89%	Yes	No
cooking spray	7.85	30	0.0460	1	33	4.18%	No	Yes
radish	9.30	14	0.0371	4	39	1.68%	No	Yes
oregano	9.45	12	0.0366	6	42	3.08%	No	Yes
smoked paprika	8.27	26	0.0298	7	38	1.94%	No	Yes
black beans	8.95	21	0.0287	8	50	12.60%	No	Yes
sesame seed	9.13	17	0.0286	9	23	0.70%	No	Yes
fat free less sodium chicken broth	5.39	70	0.0282	10	21	0.92%	No	Yes

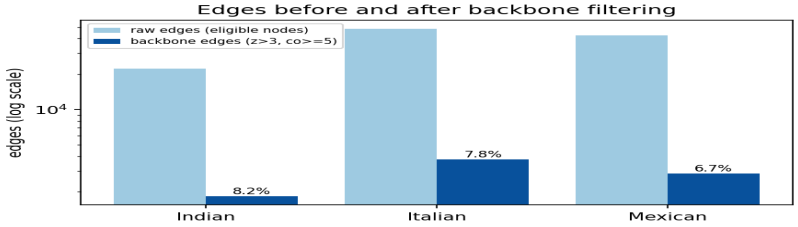


Figure 1: Raw vs statistically validated backbone edge counts (log scale) and retention percentages across cuisines.

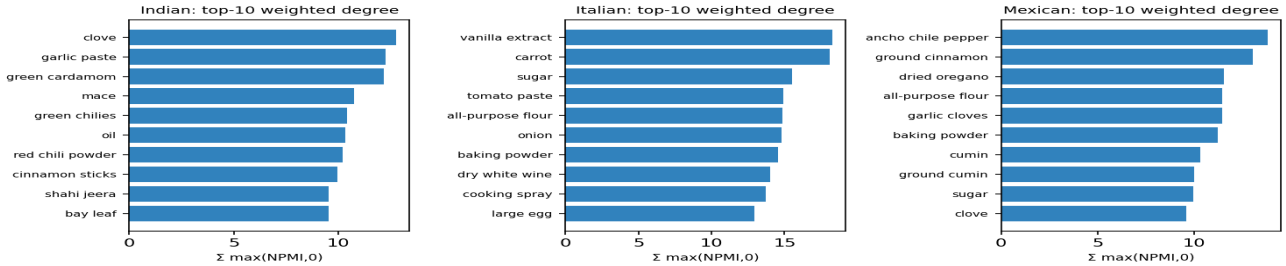


Figure 2: Top-10 weighted-degree hubs (sum of nonnegative NPMI on incident backbone edges) for Indian, Italian, and Mexican cuisines.

(b) Universal and cuisine-specific hubs

Across the union of 26 distinct top-10 weighted-degree hubs from the three cuisines, 0 ingredients qualify as universal hubs (top-10 in all three backbones), 4 qualify as two-cuisine hubs, and 22 are cuisine-specific hubs appearing in the top 10 of exactly one cuisine (Table 5, Figure 3). The 4 two-cuisine hubs are clove (top-10 in indian & mexican, ranks Indian=1, Italian=-, Mexican=10); all-purpose flour (top-10 in italian & mexican, ranks Indian=94, Italian=5, Mexican=4); baking powder (top-10 in italian & mexican, ranks Indian=74, Italian=7, Mexican=6); sugar (top-10 in italian & mexican, ranks Indian=96, Italian=3, Mexican=9) (Table 5, Figure 3).

Crucially, absence from a cuisine's top-10 weighted-degree hubs—or even absence from a backbone—does not imply culinary absence from recipes: for example, onion is a top-10 weighted-degree hub only in Italian (rank 6, strength 14.80, prevalence 15.82%), yet onion occurs in 39.79% of Indian recipes (backbone strength rank 36, strength 5.87) and 23.19% of Mexican recipes (backbone strength rank 57, strength 6.06) because ubiquitous staples co-occur broadly across many recipe types rather than concentrating into high-NPMI specialist cliques (Table 5).

Table 5: Classification of All Top-10 Weighted-Degree Hubs Across Cuisines

Ingredient	Class	Top-10 In	Rank (IN/IT/MX)	Strength (IN/IT/MX)	Prevalence % (IN/IT/MX)
clove	two-cuisine	indian,mexican	1/-/10	12.8/-/9.6	10.2/0.2/0.9
all-purpose flour	two-cuisine	italian,mexican	94/5/4	3.8/14.8/11.5	5.0/11.7/5.1
baking powder	two-cuisine	italian,mexican	74/7/6	4.4/14.5/11.2	1.6/2.4/2.7
sugar	two-cuisine	italian,mexican	96/3/9	3.7/15.5/10.0	11.2/9.7/5.3
bay leaf	cuisine-specific	indian	10/47/101	9.5/8.2/4.2	3.9/1.7/1.2
cinnamon sticks	cuisine-specific	indian	8/-/27	10.0/-/8.3	8.5/0.2/1.0
garlic paste	cuisine-specific	indian	2/-/-	12.3/-/-	7.4/0.0/0.0
green cardamom	cuisine-specific	indian	3/-/-	12.2/-/-	2.6/0.0/0.0
green chilies	cuisine-specific	indian	5/-/208	10.4/-/2.2	16.1/0.0/2.6
mace	cuisine-specific	indian	4/-/-	10.7/-/-	1.4/0.0/0.0
oil	cuisine-specific	indian	6/433/205	10.4/0.4/2.3	18.2/1.2/2.8

Ingredient	Class	Top-10 In	Rank (IN/IT/MX)	Strength (IN/IT/MX)	Prevalence % (IN/IT/MX)
red chili powder	cuisine-specific	indian	7/-/-	10.2/-/-	5.7/0.0/0.1
shahi jeera	cuisine-specific	indian	9/-/-	9.5/-/-	0.7/0.0/0.0
carrot	cuisine-specific	italian	54/2/84	5.1/18.1/4.9	6.6/4.8/2.4
cooking spray	cuisine-specific	italian	50/9/30	5.3/13.7/7.8	2.1/6.3/4.2
dry white wine	cuisine-specific	italian	-/8/-	-/14.0/-	0.3/8.4/0.2
large egg	cuisine-specific	italian	243/10/28	0.5/12.9/8.2	0.9/8.0/2.9
onion	cuisine-specific	italian	36/6/57	5.9/14.8/6.1	39.8/15.8/23.2
tomato paste	cuisine-specific	italian	58/4/102	4.9/14.9/4.2	5.3/4.8/2.3
vanilla extract	cuisine-specific	italian	-/1/13	-/18.2/9.3	0.4/2.8/1.3
ancho chile pepper	cuisine-specific	mexican	-/-/1	-/-/13.8	0.0/0.0/2.1
cumin	cuisine-specific	mexican	57/-/7	5.0/-/10.3	5.9/0.0/9.7
dried oregano	cuisine-specific	mexican	-/16/3	-/11.8/11.5	0.0/8.0/8.0
garlic cloves	cuisine-specific	mexican	28/23/5	6.5/10.5/11.4	14.9/20.7/14.3
ground cinnamon	cuisine-specific	mexican	64/40/2	4.7/8.9/13.0	4.1/1.1/2.7
ground cumin	cuisine-specific	mexican	72/346/8	4.5/1.2/10.0	22.7/0.4/20.9

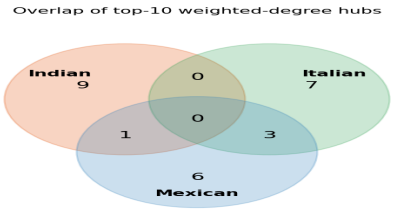


Figure 3: Overlap of top-10 weighted-degree hubs across Indian, Italian, and Mexican backbones.

(c) Louvain communities

Louvain partitioning on nonnegative NPMI weights in Indian identifies 7 communities (including 0 singleton communities on disconnected dyads/isolates) with weighted modularity Q = 0.488, led by C0 (size 61, label 'peeled fresh ginger / chopped cilantro fresh / cooking spray'); C1 (size 49, label 'heavy cream / boneless skinless chicken breast / cayenne pepper'); C2 (size 42, label 'garlic paste / oil / coriander powder') (Table 1, Table 6, Figure 4).

Louvain partitioning on nonnegative NPMI weights in Italian identifies 6 communities (including 0 singleton communities on disconnected dyads/isolates) with weighted modularity Q = 0.473, led by C0 (size 125, label 'dry white wine / arborio rice / shallot'); C1 (size 89, label 'carrot / onion / tomato paste'); C2 (size 86, label 'shredded mozzarella cheese / lasagna noodles / tomato sauce') (Table 1, Table 6, Figure 4).

Louvain partitioning on nonnegative NPMI weights in Mexican identifies 7 communities (including 0 singleton communities on disconnected dyads/isolates) with weighted modularity Q = 0.497, led by C0 (size 117, label 'lime / cumin / smoked paprika'); C1 (size 84, label 'shredded cheddar cheese / sour cream / ground beef'); C2 (size 59, label 'dried oregano / ground cloves / bay leaves') (Table 1, Table 6, Figure 4).

All community labels in Table 6 are automatically generated from the top three ingredients by within-community weighted degree and are explicitly reported as data-derived descriptive labels rather than validated semantic classifications (Table 6, Figure 4).

Table 6: Louvain Communities and Top-5 Intra-Community Members (Data-Derived Descriptive Labels)

Cuisine	Comm. ID	Size	Data-Derived Label (Top-3)	Top-5 Ingredients by Intra-Community Strength
Indian	0	61	peeled fresh ginger / chopped cilantro fresh / cooking spray	peeled fresh ginger (7.38), chopped cilantro fresh (4.91), cooking spray (4.79), garlic cloves (4.37), fresh lemon juice (4.32)
Indian	1	49	heavy cream / boneless skinless chicken breast / cayenne pepper	heavy cream (5.04), boneless skinless chicken breast (5.00), cayenne pepper (4.28), paprika (4.10), fresh ginger (4.08)
Indian	2	42	garlic paste / oil / coriander powder	garlic paste (6.13), oil (5.79), coriander powder (5.77), kasuri methi (5.40), chili powder (4.99)
Indian	3	40	mace / clove / green cardamom	mace (9.63), clove (9.56), green cardamom (8.66), cinnamon sticks (8.20), shahi jeera (7.31)
Indian	4	30	curry leaves / mustard seed / fresh curry leaves	curry leaves (4.81), mustard seed (4.33), fresh curry leaves (3.64), asafoetida powder (3.49), urad dal (3.49)
Indian	5	21	carrot / frozen peas / potato	carrot (2.81), frozen peas (2.51), potato (1.78), cauliflower (1.76), chopped cilantro (1.38)
Indian	6	18	baking powder / active dry yeast / melted butter	baking powder (4.08), active dry yeast (3.85), melted butter (3.52), all-purpose flour (3.46), baking soda (3.19)
Italian	0	125	dry white wine / arborio rice / shallot	dry white wine (10.87), arborio rice (8.94), shallot (6.64), extra-virgin olive oil (6.02), garlic cloves (5.96)
Italian	1	89	carrot / onion / tomato paste	carrot (13.77), onion (8.78), tomato paste (8.51), celery (8.26), bay leaf (7.15)
Italian	2	86	shredded mozzarella cheese / lasagna noodles / tomato sauce	shredded mozzarella cheese (9.90), lasagna noodles (7.87), tomato sauce (7.78), pasta sauce (7.61), ricotta cheese (7.36)
Italian	3	81	vanilla extract / sugar / baking powder	vanilla extract (17.21), sugar (13.45), baking powder (13.03), confectioners sugar (11.85), unsweetened cocoa powder (10.97)
Italian	4	65	purple onion / red wine vinegar / balsamic vinegar	purple onion (5.19), red wine vinegar (4.78), balsamic vinegar (3.58), red bell pepper (3.12), dijon mustard (2.71)
Italian	5	27	warm water / cooking spray / dry yeast	warm water (4.78), cooking spray (4.60), dry yeast (3.85), active dry yeast (3.72), fresh parmesan cheese (3.00)
Mexican	0	117	lime / cumin / smoked paprika	lime (6.81), cumin (6.34), smoked paprika (6.31), black beans (6.07), chili powder (5.82)
Mexican	1	84	shredded cheddar cheese / sour cream / ground beef	shredded cheddar cheese (7.07), sour cream (7.02), ground beef (6.71), taco seasoning mix (5.62), refried beans (5.35)
Mexican	2	59	dried oregano / ground cloves / bay leaves	dried oregano (6.22), ground cloves (6.18), bay leaves (6.06), garlic (4.61), apple cider vinegar (4.56)
Mexican	3	52	black peppercorn / ancho chile pepper / sesame seed	black peppercorn (7.85), ancho chile pepper (7.74), sesame seed (7.31), clove (7.28), white onion (6.07)
Mexican	4	46	all-purpose flour / baking powder / vanilla extract	all-purpose flour (10.29), baking powder (9.80), vanilla extract (8.70), ground cinnamon (8.20), unsalted butter (7.85)
Mexican	5	27	cooking spray / non-fat sour cream / chopped onion	cooking spray (4.67), non-fat sour cream (3.75), chopped onion (3.30), vegetable oil cooking spray (2.70), fat free less sodium chicken broth (2.15)
Mexican	6	23	tequila / fresh lime juice / triple sec	tequila (3.11), fresh lime juice (2.94), triple sec (2.03), honey (1.85), extra-virgin olive oil (1.82)

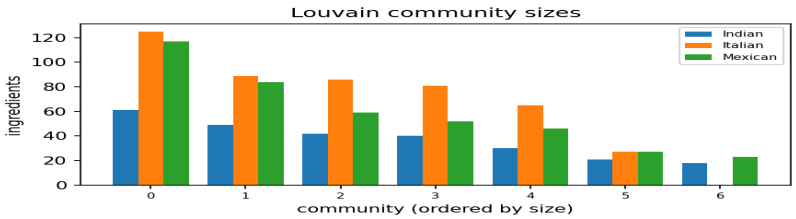


Figure 4: Louvain community size distributions across Indian, Italian, and Mexican backbones.

(d) Strongest bridge ingredients

Between Indian and Italian (124 shared backbone nodes eligible as bridge candidates), the top three structural bridges by Bridge(a; A,B) = min(Strength_A(a), Strength_B(a)) are bay leaf (rank 1, bridge score 8.21; strengths 9.54 vs 8.21; prevalences 3.93% vs 1.66%), heavy cream (rank 2, bridge score 6.79), and garlic cloves (rank 3, bridge score 6.52) (Table 7, Figure 5).

Between Indian and Mexican (134 shared backbone nodes eligible as bridge candidates), the top three structural bridges by Bridge(a; A,B) = min(Strength_A(a), Strength_B(a)) are clove (rank 1, bridge score 9.60; strengths 12.77 vs 9.60; prevalences 10.19% vs 0.89%), cinnamon sticks (rank 2, bridge score 8.26), and black peppercorn (rank 3, bridge score 7.79) (Table 7, Figure 5).

Between Italian and Mexican (195 shared backbone nodes eligible as bridge candidates), the top three structural bridges by Bridge(a; A,B) = min(Strength_A(a), Strength_B(a)) are dried oregano (rank 1, bridge score 11.51; strengths 11.83 vs 11.51; prevalences 7.99% vs 8.05%), all-purpose flour (rank 2, bridge score 11.46), and baking powder (rank 3, bridge score 11.22) (Table 7, Figure 5).

These bridge scores quantify shared structural centrality across pairs of validated ingredient backbones and do not imply historical cultural exchange, shared culinary ancestry, or causal influence (Table 7, Figure 5).

Table 7: Top-5 Structural Bridge Ingredients per Cuisine Pair

Cuisine Pair	Rank	Ingredient	Strength A	Strength B	Bridge Score	Prev. A (%)	Prev. B (%)
Indian-Italian	1	bay leaf	9.54	8.21	8.21	3.93%	1.66%
Indian-Italian	2	heavy cream	6.79	7.04	6.79	3.50%	3.83%
Indian-Italian	3	garlic cloves	6.52	10.54	6.52	14.85%	20.66%
Indian-Italian	4	garlic	6.37	10.26	6.37	24.18%	18.77%
Indian-Italian	5	onion	5.87	14.80	5.87	39.79%	15.82%
Indian-Mexican	1	clove	12.77	9.60	9.60	10.19%	0.89%
Indian-Mexican	2	cinnamon sticks	9.97	8.26	8.26	8.52%	1.04%
Indian-Mexican	3	black peppercorn	7.79	9.07	7.79	4.06%	0.78%
Indian-Mexican	4	cinnamon	7.70	7.24	7.24	5.19%	1.10%
Indian-Mexican	5	garlic cloves	6.52	11.43	6.52	14.85%	14.31%
Italian-Mexican	1	dried oregano	11.83	11.51	11.51	7.99%	8.05%
Italian-Mexican	2	all-purpose flour	14.82	11.46	11.46	11.71%	5.13%
Italian-Mexican	3	baking powder	14.53	11.22	11.22	2.37%	2.66%
Italian-Mexican	4	garlic cloves	10.54	11.43	10.54	20.66%	14.31%
Italian-Mexican	5	sugar	15.50	9.95	9.95	9.70%	5.33%

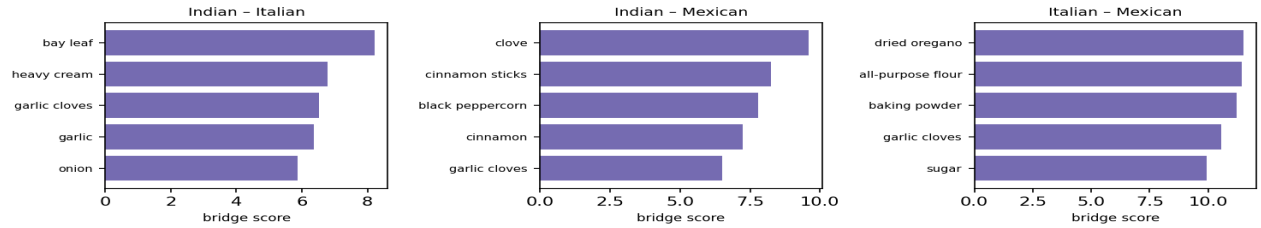


Figure 5: Top-5 bridge ingredients per cuisine pair ranked by Bridge(a; A,B) = min(Strength_A(a), Strength_B(a)).